

Jaeseung Heo

Graduate School of Artificial Intelligence
POSTECH (Pohang University of Science and Technology)

✉ jsheo12304@postech.ac.kr
🌐 jason9464.github.io

EDUCATION

POSTECH

Integrated M.S./Ph.D. Program, Graduate School of Artificial Intelligence
Advisor: Prof. Dongwoo Kim

Pohang, South Korea

Mar, 2021 - Present

Expected Graduation: Feb, 2027

Sungkyunkwan University

B.S. in Computer Science and Engineering
B.S. in Systems Management Engineering (Double Major)

Suwon, South Korea

Mar, 2016 - Aug, 2020

RESEARCH EXPERIENCE

MATS Research

Exploration Phase, Neel Nanda Stream

Sep, 2026 - Present

RESEARCH INTERESTS

- **Training data attribution for LLMs:** Developing attribution methods applicable to LLMs, with the goal of scaling them to models with over 100B parameters.
- **The science of pre- and post-training:** Understanding which training data gives rise to which model behaviors, and whether filtering or perturbing the data can steer a model toward desired behaviors.
- **Mechanistic interpretability:** Explaining the mechanisms behind LLM behavior, with tools such as the Jacobian lens, sparse autoencoders, and cross-layer transcoders.

PUBLICATIONS

- **Scaling Influence Functions in LLMs through Eigenbasis-Corrected One-Bit Gradient Projection** *arXiv '26*
Jaeseung Heo, J Rosser, Dongwoo Kim
- **Interaction-Aware Influence Functions for Group Attribution** *ICML '26*
Jaeseung Heo, Kyeongheung Yun, Youngbin Choi, Sehyun Hwang, Jungseul Ok, Dongwoo Kim *MI Workshop*
- **Posterior Label Smoothing for Node Classification** *AAAI '26*
Jaeseung Heo, MoonJeong Park, Dongwoo Kim *Oral Pres.*
- **Long Live the Librarian! A Persistent Search Sub-Agent for Energy-Efficient Multi-Agent Software Engineering Systems** *EMNLP '26*
Seunghyuk Cho, Sunghyun Choi, Jaeseung Heo, Youngbin Choi, Saemi Moon, MoonJeong Park, Dongwoo Kim *Oral Pres.*
- **Transductive Generalization and Its Application to Graph Node Classification** *NeurIPS '26*
MoonJeong Park*, Seungbeom Lee*, Kyungmin Kim, Jaeseung Heo, Seunghyuk Cho, Shouheng Li, Sangdon Park, Dongwoo Kim
- **Influence Functions for Edge Edits in Non-Convex Graph Neural Networks** *NeurIPS '25*
Jaeseung Heo, Kyeongheung Yun, Seokwon Yoon, MoonJeong Park, Jungseul Ok, Dongwoo Kim
- **The Oversmoothing Fallacy: A Misguided Narrative in GNN Research** *arXiv '25*
MoonJeong Park, Sunghyun Choi, Jaeseung Heo, Eunhyeok Park, Dongwoo Kim
- **EPIC: Graph Augmentation with Edit Path Interpolation via Learnable Cost** *IJCAI '24*
Jaeseung Heo*, Seungbeom Lee*, Sungsoo Ahn, Dongwoo Kim

*Equal contribution.

HONORS AND AWARDS

- BK21 Outstanding Research Paper Award, POSTECH
- 2025, 2026
- Dean’s List, Sungkyunkwan University
- 2018, 2019, 2020

SERVICE AND TEACHING

- Reviewer, ICLR, ICML, ICML MI Workshop, NeurIPS
- Teaching Assistant, CSED700H: Convex Optimization, POSTECH
- Fall 2022
- Lecturer, Youth AI & Big Data Academy, AI Expert Training Course, POSCO
- 2021, 2022

REFERENCES

Dongwoo Kim Associate Professor POSTECH dongwoo.kim@postech.ac.kr	Jungseul Ok Associate Professor POSTECH jungseul.ok@postech.ac.kr	MoonJeong Park Ph.D. Candidate POSTECH mjeongp@postech.ac.kr
---	---	--